Biomedical Text Embeddings Weakly Capture Chromatin Accessibility Similarity: A Large-Scale Measurement Study

***Abstract*—The extent to which biomedical text metadata can serve as a proxy for chromatin accessibility similarity remains unknown. We conducted a systematic measurement study comparing pairwise cosine similarity matrices across 12 distinct embedding types (Atacformer, regulatory potential (RP), four biomedical language models tested in full-metadata and cell-type-name-only configurations, term frequency-inverse document frequency (TF-IDF), and a random Gaussian baseline) across 32,245 human assay for transposase-accessible chromatin with sequencing (ATAC-seq) experiments from ChIP-Atlas. Using Spearman rank correlation, Mantel permutation tests, and top-k neighborhood overlap, we find that apparent text–chromatin agreement is driven largely by experiments with few called peaks. These experiments retain mainly promoter peaks (median 69 to 75% within 1 kb of a transcription start site) and therefore resemble one another in both chromatin representations regardless of cell type. On the full dataset, text embeddings correlate with Atacformer similarity at $r = 0.20$ to 0.28, but among the 17,232 well-covered experiments ($\geq$5,000 peaks) this falls to $r = 0.011$ to 0.016, with less than 0.1% shared nearest neighbors. Text embeddings instead show weak but consistent agreement with RP ($r = 0.06$ to 0.10; 31 to 35% of top-5 neighbors shared), reflecting the coarse cell-type structure that both encode. Sample titles, which encode data provenance rather than biology, carry most of the full-dataset text signal. TF-IDF matches all large language models (LLMs) in well-covered experiments, indicating that semantic understanding adds little beyond surface-level vocabulary. These results bound the utility of text metadata as a proxy for chromatin similarity and show that peak count must be controlled when comparing chromatin representations across public repositories.**

***Index Terms*—genomics, bioinformatics, chromatin, natural language processing, transformers, feature extraction, correlation, similarity measures**

I. Introduction

The rapid growth of epigenomics databases such as ChIP-Atlas [1], Encyclopedia of DNA Elements (ENCODE) [2], and Roadmap Epigenomics [3] has produced tens of thousands of ATAC-seq experiments across diverse human cell types and conditions. Organizing, retrieving, and comparing these experiments requires similarity measures that reflect the underlying biology. A natural question arises: to what extent can text metadata associated with experiments (cell type labels, experimental descriptions, and publication titles) serve as a proxy for chromatin accessibility similarity?

This question has direct practical implications. If text embeddings of experimental metadata closely approximate chromatin-based similarity measures (such as those derived from Atacformer or RP), they could enable natural-language retrieval and annotation of ATAC-seq data without access to the raw signal, potentially democratizing the use of large epigenomics repositories. If agreement is limited, the results instead point to fundamental limits on what text can capture and motivate investment in chromatin-specific representation learning.

Recent advances in biomedical natural language processing have produced powerful text encoders pretrained on large biomedical corpora. PubMedBERT [4] is pretrained from scratch on PubMed abstracts and full-text articles. SapBERT [5] and MedCPT [6] are both initialized from a pretrained PubMedBERT checkpoint and further fine-tuned for different objectives: SapBERT via self-alignment on Unified Medical Language System (UMLS) synonym pairs, and MedCPT via contrastive learning on PubMed search logs. These models achieve strong performance on biomedical entity recognition, relation extraction, and information retrieval, yet whether their semantic representations capture the biological relationships encoded in chromatin accessibility data has not been systematically evaluated.

Atacformer [7] is a transformer trained on ATAC-seq data using triplet margin loss to produce experiment-level embeddings that cluster by cell type and chromatin state. Regulatory potential (RP) [8] provides a complementary, functional-genomics-derived signal: a gene-activation-weighted summary of chromatin accessibility indexed by gene transcription start sites (TSS).

We compare pairwise cosine similarity matrices from 12 embedding types (chromatin models, biomedical LLMs in full-metadata and name-only configurations, TF-IDF, and a random baseline) across 32,245 curated human ATAC-seq experiments, treating both agreement and disagreement as informative findings. Because public repositories mix experiments of very different quality, we also examine how the number of called peaks shapes these comparisons, and base our primary analysis on well-covered experiments.

The key findings are:

- Peak count dominates similarity in both chromatin representations; low-peak experiments, which retain mainly promoter peaks, inflate apparent text–chromatin agreement on the full dataset.
- In well-covered experiments, text does not track Atacformer similarity and tracks RP only weakly, capturing coarse cell-type identity rather than fine-grained chromatin variation.

- Vocabulary overlap alone (TF-IDF) matches all biomedical LLMs, and sample titles, which encode data provenance, carry most of the full-dataset text signal.

II. Methods

Fig. 1 summarizes the overall study design: chromatin-based embeddings (Atacformer, RP) and text-based embeddings (biomedical and general language models, TF-IDF, random) are computed for each experiment, converted into pairwise similarity matrices at two levels of aggregation, and compared using the four metrics described in Section II-E.

A. Dataset

We obtained 48,822 human ATAC-seq experiments from ChIP-Atlas (hg38 assembly) [1]. We removed 16,567 experiments with corrupted or empty BED files and 10 experiments producing zero regulatory potential signal near gene TSSs, yielding a curated set of 32,245 experiments spanning 1,108 cell types.

ChIP-Atlas peak sets are called at a stringent threshold ($q < 10^{-10}$), and the number of peaks per experiment varies over more than three orders of magnitude (median 6,287; interquartile range 780–19,836; 8,964 experiments have fewer than 1,000 peaks). Because experiments with few peaks retain mainly the strongest, largely constitutive sites (Section III-A), our primary analysis uses the 17,232 experiments with at least 5,000 peaks (hereafter *well-covered* experiments). We report results on all 32,245 experiments as a contrast.

B. Atacformer Embeddings

We encoded each experiment using databio/atacformer-base-hg38 (AtacformerForCellClustering) [7], a transformer trained with triplet margin loss. Peak BED files were tokenized using the gtars Tokenizer with a vocabulary of 890K genomic regions. Up to 1,024 randomly sampled peak tokens were mean-pooled to produce a 192-dimensional embedding per experiment.

C. Regulatory Potential Embeddings

We computed RP for each experiment using a LISA2-style exponential decay model [8]: each chromatin peak's contribution to a given gene is weighted by $w(d) = e^{-d/10\,\mathrm{kb}}$, where d is the distance to the gene's TSS, with peaks overlapping annotated exons assigned a fixed weight of 1.0 (replacing the exponential decay for those peaks). Summing across all peaks yields a per-gene RP score; stacking over all hg38 autosomal gene TSSs maps each experiment to a 40,772-dimensional vector.

D. Text Embeddings

This section describes text- and metadata-based embedding methods only; chromatin-based embeddings (Atacformer and RP) are described in the two preceding subsections.

For each experiment we constructed a full metadata string by concatenating available fields: {cell_type}. {cell_type_class}. {cell_type_description}. {title}. We embedded this string using four encoders, all of which produce 768-dimensional embeddings: PubMedBERT [4] (microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext, mean pooling), SapBERT [5] (cambridgeltl/

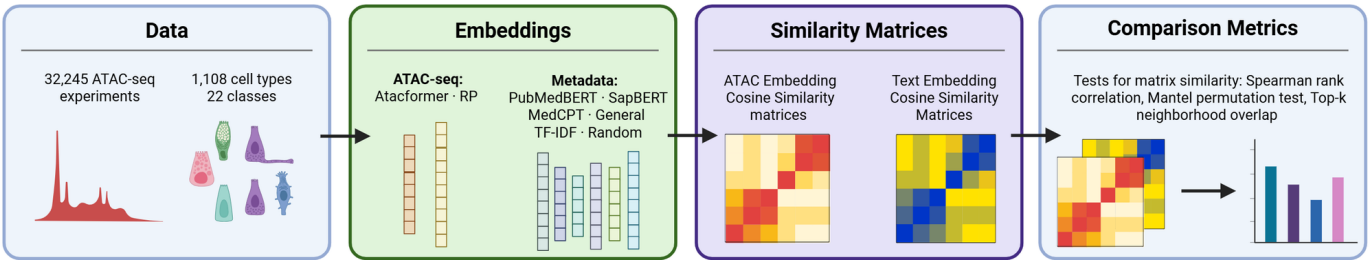


Fig. 1. Study design. ATAC-seq experiments are embedded using chromatin-based models (Atacformer, RP), text encoders (biomedical and general language models, TF-IDF), and a random baseline. Pairwise cosine similarity matrices are computed at cell-type-aggregated and full-dataset levels and compared using four complementary metrics. Created with BioRender.com [9].

SapBERT-from-PubMedBERT-fulltext, classification token (CLS) pooling), MedCPT [6] (ncbi/MedCPT-Query-Encoder, mean pooling), and a General encoder [10] (sentence-transformers/all-mpnet-base-v2, mean pooling). To isolate the contribution of cell type identity, we also encoded each experiment using only the cell type name string with all four models (cell-type-name-only ablation). We additionally computed TF-IDF embeddings (top 2,048 features, sublinear TF, 2,048-dimensional) and a random Gaussian baseline (192-dimensional, seed 42).

E. Similarity Matrices and Comparison Metrics

For each embedding type we constructed pairwise cosine similarity matrices at two levels of aggregation.

Cell-type-aggregated level. Individual experiment embeddings were mean-pooled within each cell type to produce one representative vector per cell type. A corpus mean vector was computed as the arithmetic mean over up to 10,000 randomly sampled individual experiment embeddings and subtracted from each cell-type mean vector. For every pair of cell types (i, j), the cosine similarity between their centered vectors was computed and stored in entry (i, j) of the matrix, yielding a symmetric $1{,}108 \times 1{,}108$ matrix with ones on the diagonal.

Full-dataset level. The same corpus-centering procedure was applied to all 32,245 individual experiment embeddings. For every pair of experiments (i, j), the cosine similarity between their centered embedding vectors was computed and stored in entry (i, j), yielding a symmetric $32{,}245 \times 32{,}245$ matrix with ones on the diagonal.

Corpus centering corrects for anisotropy in transformer mean-pool representations: without it, embeddings tend to cluster in a narrow cone in the embedding space, which inflates cosine similarities uniformly and obscures relative differences in pairwise structure. The centering vector was estimated from individual experiments (not from cell-type means) to avoid conflating aggregation level with the centering baseline.

We quantified agreement between each text-embedding similarity matrix and each chromatin reference (Atacformer and RP) using four metrics:

1) Spearman rank correlation (r) on upper-triangle edge weights.
2) Mantel permutation test (1,000 permutations) for empirical p-values at the cell-type level.
3) Bootstrap confidence intervals for r (1,000 resamples of cell types/experiments, with 50,000 randomly sampled pairs per resample), reported at both the cell-type and full-dataset scale as a scale-independent measure of effect-size precision.
4) Top-k neighborhood overlap: mean fraction of shared top-k nearest neighbors, $k \in \{5, 10, 20, 50\}$.

All analyses were conducted with both Atacformer and RP as reference structures (i.e., the chromatin similarity structures against which text-based embeddings are evaluated). For the primary (well-covered) analysis, cell-type mean vectors were recomputed from well-covered experiments only, and experiment-level matrices were restricted to the 17,232 well-covered experiments; the corpus mean used for centering was unchanged. Full-scale correlations were estimated on 2×10^6 randomly sampled pairs.

F. Technical Covariates and Sensitivity Analyses

For each experiment we recorded the number of peaks in its ChIP-Atlas BED file and the fraction of peaks whose midpoint lies within 1 kb of an annotated RefSeq TSS (the same hg38 annotation used for RP). For a pair of experiments we summarized peak count by *depth match*, $-|\log_{10} n_i - \log_{10} n_j|$, and *mean depth*, $(\log_{10} n_i + \log_{10} n_j)/2$. We used these quantities in four analyses: (i) Spearman correlations computed within peak-count strata (both experiments with $<$1K, 1–5K, 5–10K, or $\geq$10K peaks); (ii) partial Spearman correlations controlling for depth match and mean depth (and, separately, for same-cell-type and same-cell-type-class indicators), computed by regressing covariate ranks out of each variable's ranks; (iii) a title-only TF-IDF embedding (same settings as above, fit on the {title} field alone) to isolate the contribution of sample titles; and (iv) recomputation of all similarities without corpus centering. Uncertainty for partial correlations was estimated by an experiment-level block bootstrap (1,000 resamples of experiments, 200,000 pairs each).

G. Decomposing LLM Signal from TF-IDF

To determine whether LLMs encode biological information beyond surface-level vocabulary overlap, we conducted three complementary analyses comparing the full-text LLM embeddings against TF-IDF, both in well-covered experiments and on the full 32,245-experiment dataset.

Pairwise similarity scatter. For each LLM, we sampled up to 25,000 random experiment pairs and plotted TF-IDF pairwise similarity against LLM pairwise similarity, coloring each point by Atacformer similarity (Fig. 6). Points above the diagonal indicate pairs where the LLM assigns higher similarity than TF-IDF; points below indicate the reverse. The Spearman correlation between TF-IDF and LLM similarity vectors was computed over the same sample.

Partial Spearman correlation. We computed the partial Spearman rank correlation of each LLM similarity matrix with Atacformer, controlling for TF-IDF:

$$r_{LM,AC|TF} = \frac{r_{LM,AC} - r_{LM,TF}\, r_{AC,TF}}{\sqrt{(1 - r_{LM,TF}^2)(1 - r_{AC,TF}^2)}} \qquad (1)$$

where $r_{X,Y}$ denotes the Spearman correlation of the upper-triangle edge weights of matrices X and Y. At the full scale, correlations were estimated on a random subsample of 2×10^6 pairs from the $\sim$520M-element upper triangle. A positive partial correlation indicates that the LLM encodes biological structure beyond vocabulary overlap; a value near zero or negative indicates its Atacformer alignment is statistically accounted for by TF-IDF (i.e., once TF-IDF's shared variance with Atacformer is removed, the LLM contributes no additional association with chromatin similarity).

Cross top-k neighborhood overlap. For each LLM and each $k \in \{5, 10, 20, 50\}$, we computed the mean fraction of top-k LLM nearest neighbors that are also top-k TF-IDF nearest neighbors, using the same topk_overlap procedure described above. High overlap indicates the LLM's local structure recapitulates TF-IDF; low overlap indicates genuinely distinct neighbor assignments.

III. Results

A. Peak Count Inflates Apparent Text–Chromatin Agreement

Experiments with few peaks retain mainly promoter peaks. Below 10,000 peaks, the median experiment has 69 to 75% of its peaks within 1 kb of a TSS; this falls to 39% at 20,000–50,000 peaks and 24% above 50,000 (Fig. 2a). Promoters of broadly expressed genes are accessible in most cell types, so low-peak experiments resemble one another regardless of cell type.

This shared promoter signal dominates the similarity structure of both chromatin representations. Across all 32,245 experiments, Atacformer similarity correlated with depth match (Section II-F) at $r = 0.66$, and RP similarity at $r = 0.59$, larger than any text–chromatin correlation we observed. After controlling for depth match and mean depth, Atacformer–RP agreement fell from $r = 0.215$ to -0.018. Across 406 cell types with ≥ 10 experiments, within-cell-type Atacformer similarity was strongly predicted by the spread of peak counts within that cell type (Spearman $\rho = -0.77$ with the SD of $\log_{10}$ peak count; RP: $\rho = -0.71$).

Text–Atacformer agreement on the full dataset is concentrated in these low-peak experiments (Fig. 2b). When both experiments have fewer than 1,000 peaks, text models correlate with Atacformer at $r = 0.38$ to 0.41; this falls to 0.08–0.09 at 1,000–5,000 peaks, 0.02–0.04 at 5,000–10,000, and 0.007–0.012 above 10,000. Text–RP agreement shows the opposite trend, rising from $r = 0.04$–0.10 in the lowest stratum to 0.08–0.14 in the highest (Fig. 2c).

Within the metadata, sample titles carry most of the text signal. A TF-IDF embedding of the title field alone correlated with Atacformer as strongly as PubMedBERT on the full metadata string ($r = 0.275$ for both), and within a cell type it exceeded every full-text model ($r = 0.24$ vs. 0.20–0.22). Controlling for title similarity reduced within-cell-type text–Atacformer correlations to $r \approx 0.05$. ChIP-Atlas titles are SRA/GEO sample names (e.g., "Illumina HiSeq 2500 sequencing of SAMD00123421"), so this signal reflects data provenance (study, platform, and naming conventions), which covaries with library depth and quality, rather than biological content. Cell-type labels themselves explained little of the full-data correlation: controlling for same-cell-type and same-class indicators reduced PubMedBERT's correlation with Atacformer only from 0.275 to 0.254.

The qualitative examples in Tables II and III, which pair experiments with high chromatin similarity but low text similarity, illustrate the same artifact. All ten RP pairs consist of two low-peak experiments ($\leq$1,438 peaks each, against a corpus median of 6,287), as does the leading Atacformer pair (NCI-H1299 and CD4+ T cells; 1,181 and 1,272 peaks). We therefore base our primary analysis on well-covered experiments ($\geq$5,000 peaks) and report full-dataset results as a contrast. The 5,000-peak threshold is pragmatic: experiments with 5,000–10,000 peaks remain promoter-rich (Fig. 2a), but experiment-level correlations computed at a 10,000-peak threshold lead to the same conclusions.

B. In Well-Covered Experiments, Text Tracks RP Weakly and Atacformer Not at All

Sanity checks confirmed that Atacformer and RP cell-type means produce coherent lineage structure, with immune, neural, and cancer cell types forming distinct clusters, while text-based embeddings show coarser organization (Fig. 3).

Among the 17,232 well-covered experiments (954 cell types), text embeddings showed no positive agreement with Atacformer (Table I). At the cell-type-aggregated level, all full-text models and TF-IDF correlated slightly negatively with Atacformer ($r = -0.074$ to -0.023; Mantel $p > 0.98$). At the experiment level, correlations were indistinguishable from zero in practical terms ($r = 0.011$ to 0.016), compared with $r = 0.20$ to 0.30 on the full dataset. In contrast, all full-text models and TF-IDF showed weak but consistent agreement with RP, both at the cell-type level ($r = 0.038$ to 0.068; Mantel $p = 0.001$) and at the experiment level ($r = 0.061$ to 0.098). Atacformer and RP agreed with each other at the cell-type level ($r = 0.365$) but only modestly across experiments ($r = 0.107$).

Cell-type-name-only variants agreed with RP less than the full-text models at the cell-type level ($r = 0.003$ to 0.021), indicating that the description and title fields contribute beyond

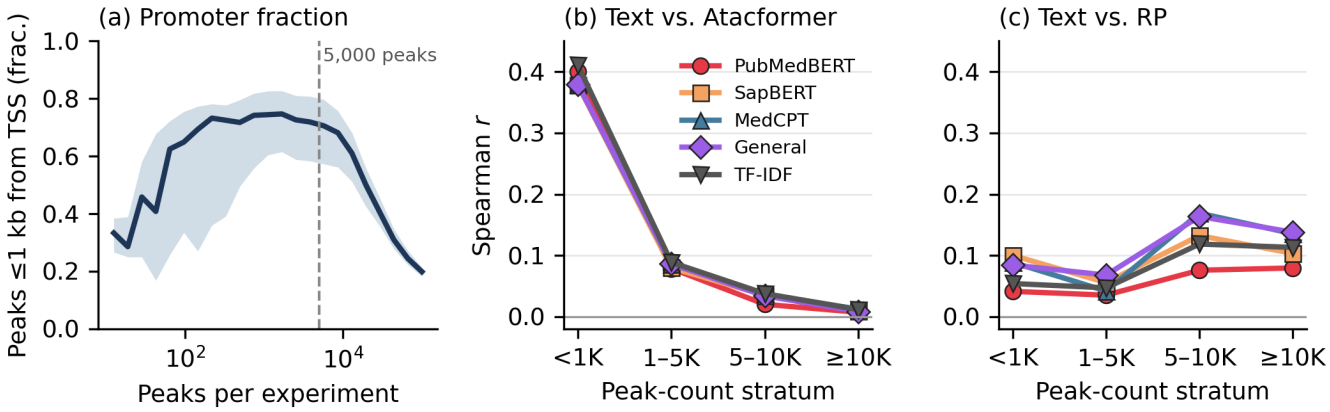


Fig. 2. Peak count and text–chromatin agreement. (a) Fraction of each experiment's peaks within 1 kb of an annotated TSS vs. number of peaks (median and interquartile range in log-spaced bins; all 32,245 experiments). Dashed line: 5,000-peak threshold for the primary analysis. (b,c) Spearman correlation of text-embedding similarity with Atacformer (b) and RP (c) similarity, computed separately for pairs in which both experiments fall in the same peak-count stratum (from 2×10^6 random pairs).

the cell-type name; at the experiment level they reached $r = 0.035$ to 0.080.

C. Biomedical Language Models Add Little Beyond Vocabulary Overlap

In well-covered experiments, TF-IDF matched the LLMs against both references (Table I). Partial correlations with Atacformer after controlling for TF-IDF were near zero for all four LLMs ($r_{\text{partial}} = 0.000$ to 0.006; bootstrap 95% CIs within $[-0.004, 0.011]$). Against RP, MedCPT, the general encoder, and SapBERT retained small residual associations beyond TF-IDF ($r_{\text{partial}} = 0.046$, 0.046, and 0.037), whereas PubMedBERT did not (0.003). LLM and TF-IDF similarity structures were strongly correlated ($r = 0.52$ to 0.73), and 44 to 45% of each LLM's top-5 neighbors were also TF-IDF neighbors.

On the full dataset (Figs. 4–7), PubMedBERT ($r_{\text{partial}} = +0.051$; 95% CI $[0.048, 0.059]$) and SapBERT ($+0.045$; $[0.039, 0.049]$) retained positive partial correlations with Atacformer, whereas MedCPT (-0.030) and the general encoder (-0.042) did not. This pattern is robust to experiment-level resampling but disappears in well-covered experiments, and it depends on corpus centering: without centering, all three biomedical models show large residuals (e.g., PubMedBERT $r_{\text{partial}} = 0.257$), and PubMedBERT ($r = 0.292$) rather than TF-IDF ($r = 0.153$) shows the highest raw correlation with Atacformer. We therefore do not interpret the full-dataset differences among LLMs as evidence about pretraining or fine-tuning objectives.

D. Text Shares Local Structure with RP but Not Atacformer

Top-k neighborhood overlap sharpened the contrast between the two references. In well-covered experiments, full-text models shared 31 to 35% of their top-5 neighbors with RP (PubMedBERT: 34.7%; SapBERT: 34.2%; General: 32.8%; MedCPT: 30.9%; TF-IDF: 28.2%; name-only models: about 12%), but less than 0.1% with Atacformer (chance: 0.03%). Atacformer and RP themselves shared only 0.1% of top-5 neighbors. On the full dataset, the corresponding values were 20 to 23% for RP and 2 to 3% for Atacformer (Figs. 8–10).

Neighborhood purity explains this dissociation. In well-covered experiments, 76% of an experiment's top-5 RP neighbors shared its cell-type label (text models: 93 to 96%), compared with 1.3% for Atacformer, barely above the 0.8% expected by chance (full dataset: RP 72%, Atacformer 21%). Within well-covered experiments, Atacformer similarity was no longer associated with depth match ($r = 0.06$), whereas RP similarity still was ($r = 0.57$). RP experiments of the same cell type were far more self-similar than Atacformer experiments (median within-cell-type similarity 0.51 vs. 0.17 across 796 cell types; Atacformer lower for 96.7%; Wilcoxon signed-rank $p = 1.1 \times 10^{-125}$; full-dataset values in Figs. 11–12).

RP therefore encodes cell-type identity that text metadata also captures, but it remains sensitive to peak count. The Atacformer embeddings used here, by contrast, carry little cell-type information for peak-rich experiments, so their low agreement with text should not be read as evidence that Atacformer captures finer biological variation that text misses. One possible explanation, which we did not test, is that each embedding summarizes at most 1,024 randomly sampled peaks (Section II-B), a small and variable fraction of the peaks in a well-covered experiment.

TABLE I

SPEARMAN RANK CORRELATION (r) OF EACH EMBEDDING'S SIMILARITY WITH ATACFORMER AND RP SIMILARITY. PRIMARY ANALYSIS: WELL-COVERED EXPERIMENTS ($\geq$5,000 PEAKS; 954 CELL TYPES, 17,232 EXPERIMENTS), WITH 95% BOOTSTRAP CONFIDENCE INTERVALS AT THE EXPERIMENT LEVEL (1,000 RESAMPLES OF EXPERIMENTS, 50,000 PAIRS EACH). CONTRAST: ALL 32,245 EXPERIMENTS, AS IN THE ORIGINAL ANALYSIS. SIGNIFICANCE MARKERS (ONE-SIDED MANTEL PERMUTATION TEST, 1,000 PERMUTATIONS) ARE REPORTED AT THE CELL-TYPE LEVEL ONLY.

	vs. Atacformer			vs. RP		
	$\geq$5,000 peaks		All	$\geq$5,000 peaks		All
Embedding	Cell type	Experiment [95% CI]	Experiment	Cell type	Experiment [95% CI]	Experiment
Atacformer / RP	0.365^{***}	0.107 [0.097, 0.116]	0.216	0.365^{***}	0.107 [0.097, 0.117]	0.216
PubMedBERT (full text)	-0.030^{ns}	0.011 [0.001, 0.021]	0.276	0.038^{***}	0.061 [0.051, 0.072]	0.092
SapBERT (full text)	-0.023^{ns}	0.013 [0.004, 0.022]	0.248	0.067^{***}	0.079 [0.069, 0.088]	0.074
MedCPT (full text)	-0.074^{ns}	0.016 [0.006, 0.025]	0.229	0.062^{***}	0.098 [0.087, 0.107]	0.090
General (full text)	-0.053^{ns}	0.014 [0.005, 0.023]	0.196	0.068^{***}	0.095 [0.085, 0.104]	0.088
TF-IDF	-0.057^{ns}	0.016 [0.007, 0.026]	0.300	0.065^{***}	0.091 [0.081, 0.101]	0.089
PubMedBERT (name only)	-0.071^{ns}	0.005 [$-$0.004, 0.013]	0.090	0.003^{ns}	0.035 [0.027, 0.044]	0.018
SapBERT (name only)	-0.101^{ns}	0.008 [$-$0.001, 0.017]	0.137	0.013^{ns}	0.066 [0.056, 0.074]	0.026
MedCPT (name only)	-0.098^{ns}	0.012 [0.002, 0.020]	0.068	0.021^{*}	0.080 [0.070, 0.089]	0.022
General (name only)	-0.060^{ns}	0.006 [$-$0.003, 0.016]	0.090	0.015^{*}	0.039 [0.030, 0.048]	0.010
Random Gaussian	0.003^{*}	0.000 [$-$0.008, 0.009]	0.000	0.003^{*}	0.000 [$-$0.009, 0.009]	0.000

$^{***}p \leq 0.001$; $^{*}p < 0.05$; ^{ns} not significant. The marginal random-baseline result arises because subtracting the corpus mean gives cell-type means of large cell types a shared direction.

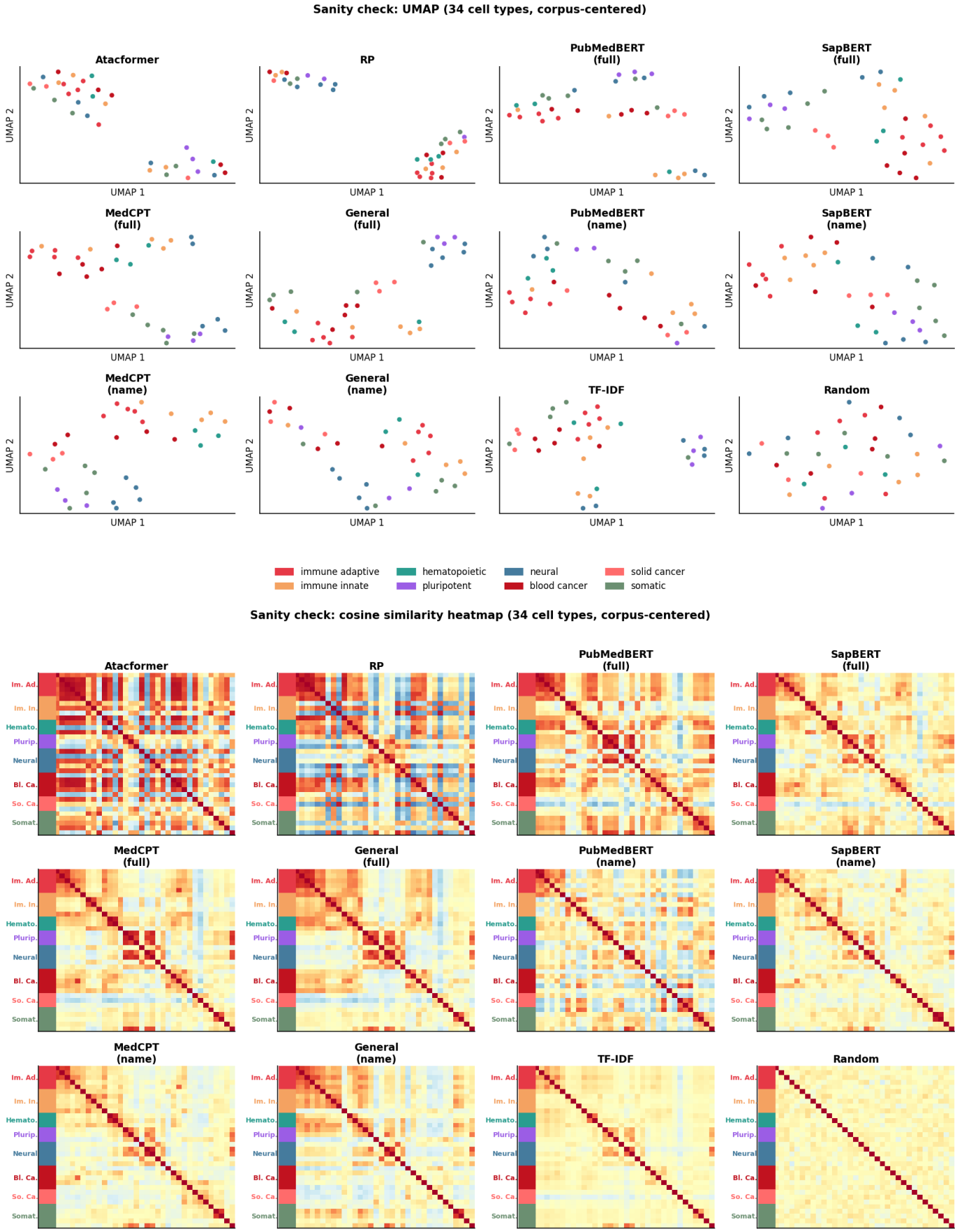


Fig. 3. Sanity check Uniform Manifold Approximation and Projection (UMAP) projections (top) and cosine similarity heatmaps (bottom) for all 12 embedding conditions over 34 representative cell types manually selected to span major lineages (immune, neural, stem/progenitor, cancer, and stromal). Atacformer and RP show biologically coherent lineage clustering; text-based models show coarser, cell-type-level organization.

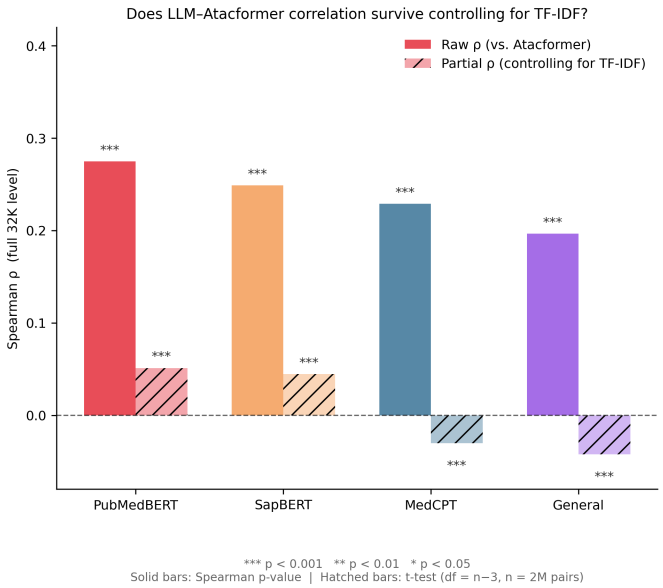


Fig. 4. All 32,245 experiments (contrast analysis). Spearman r vs. Atacformer (solid bars) and partial Spearman r_{partial} after controlling for TF-IDF (hatched bars). In well-covered experiments, all partial correlations are ≤ 0.006 (Section III-C).

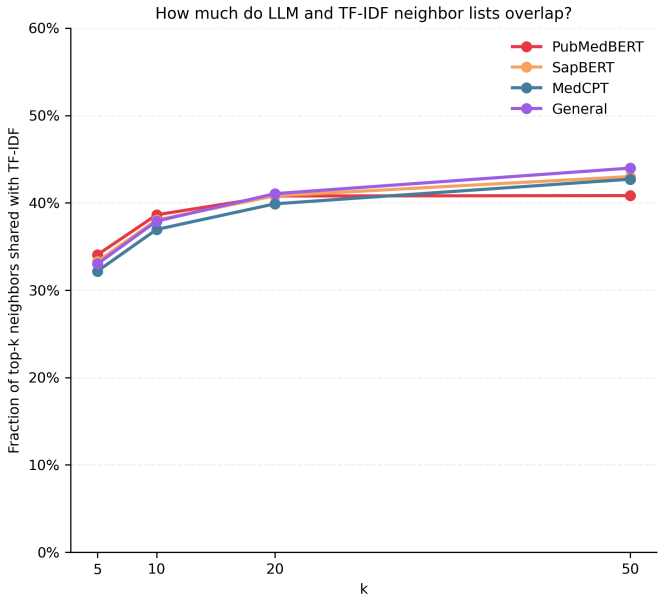


Fig. 5. All 32,245 experiments (contrast analysis). Fraction of top-k LLM nearest neighbors shared with TF-IDF ($\sim$33% at $k = 5$; 44–45% in well-covered experiments).

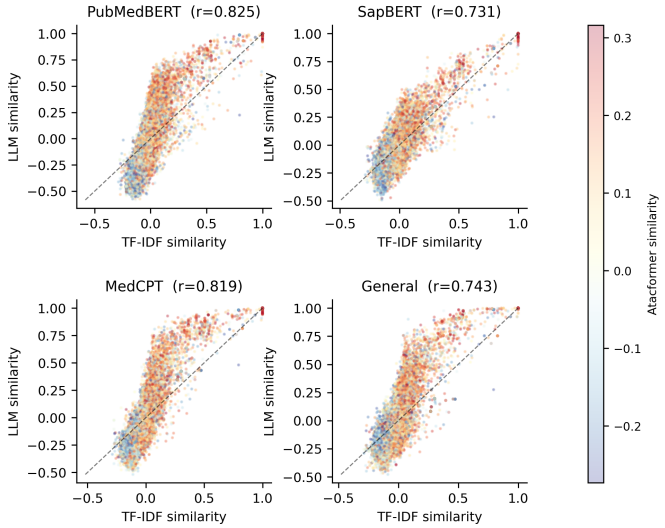


Fig. 6. All 32,245 experiments (contrast analysis). TF-IDF (x-axis) vs. LLM (y-axis) pairwise similarity for 25,000 randomly sampled experiment pairs, colored by Atacformer similarity (RdYlBu; warmer = higher). Points above the diagonal indicate pairs the LLM scores higher than TF-IDF.

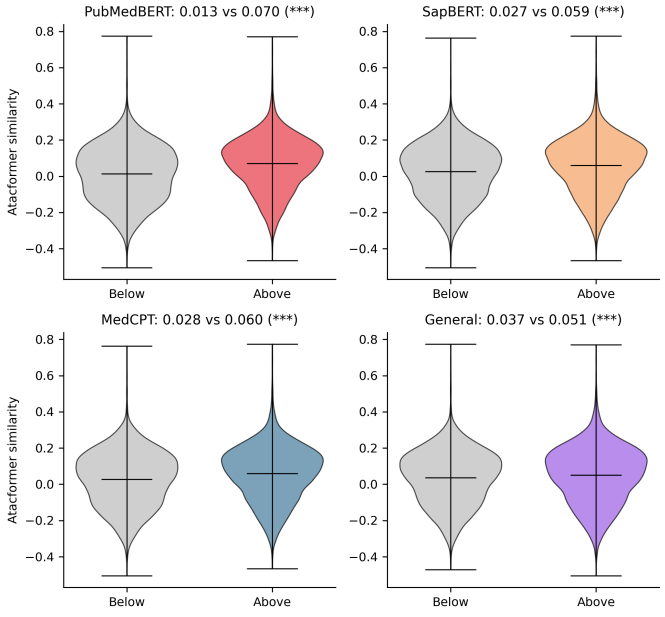


Fig. 7. All 32,245 experiments (contrast analysis). Atacformer similarity for below- vs. above-diagonal pairs in Fig. 6 (Welch's t-test). For all four models, above-diagonal pairs have higher mean Atacformer similarity (PubMedBERT: 0.070 vs. 0.013; SapBERT: 0.059 vs. 0.027; MedCPT: 0.060 vs. 0.028; General: 0.051 vs. 0.037); d denotes Cohen's d effect size (PubMedBERT: $d = 0.38$; SapBERT: $d = 0.22$; MedCPT: $d = 0.21$; General: $d = 0.09$).

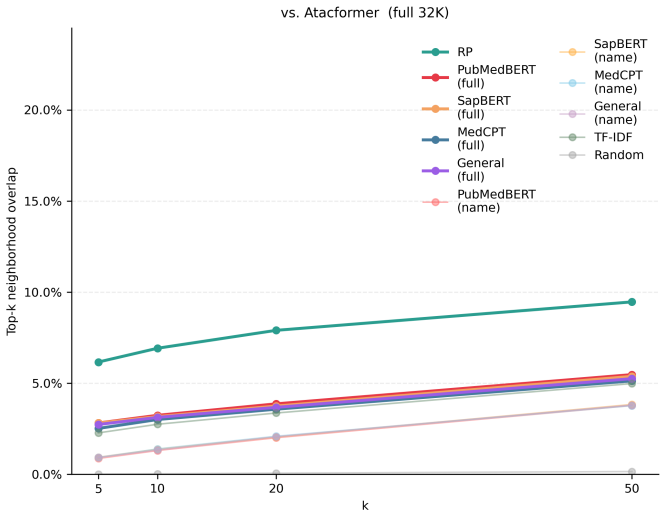


Fig. 8. All 32,245 experiments (contrast analysis). Top-k neighborhood overlap vs. Atacformer. Full-text models share $\sim$2 to 3% of nearest neighbors with Atacformer at $k = 5$ ($<$0.1% in well-covered experiments).

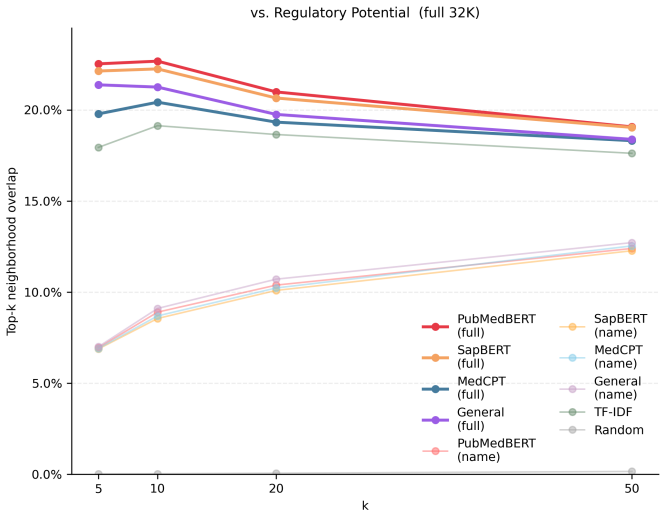


Fig. 9. All 32,245 experiments (contrast analysis). Top-k neighborhood overlap vs. regulatory potential (RP). Full-text models share $\sim$20 to 23% of nearest neighbors with RP at $k = 5$ (31 to 35% in well-covered experiments).

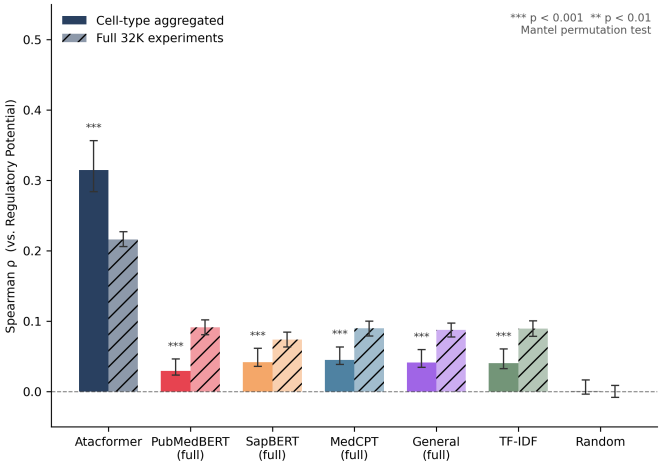


Fig. 10. All 32,245 experiments (contrast analysis). Spearman rank correlation between each embedding type and regulatory potential (RP) (name-only variants omitted for clarity). Error bars show 95% bootstrap confidence intervals; significance markers (Mantel permutation test) are shown for the cell-type-aggregated bars only. Well-covered results are in Table I.

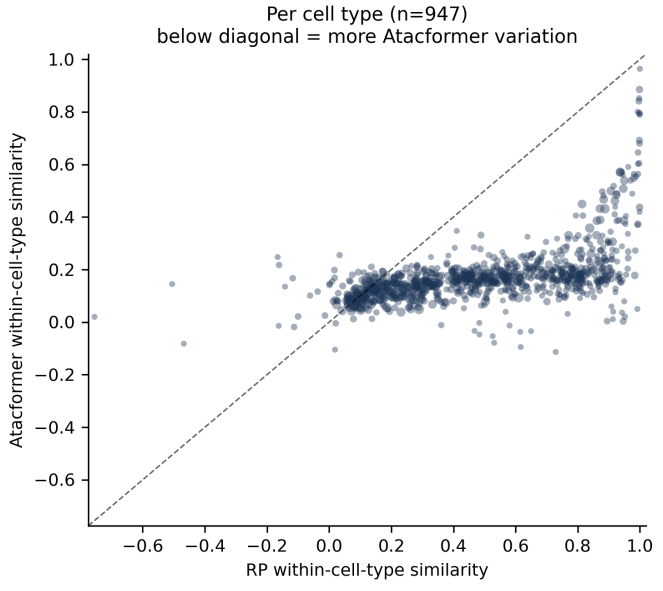


Fig. 11. All 32,245 experiments (contrast analysis). Per-cell-type within-cell-type similarity for Atacformer vs. RP, computed as mean pairwise cosine similarity among experiments sharing a cell-type label (947 cell types with ≥ 2 experiments). Points below the diagonal indicate lower within-cell-type similarity for Atacformer than RP.

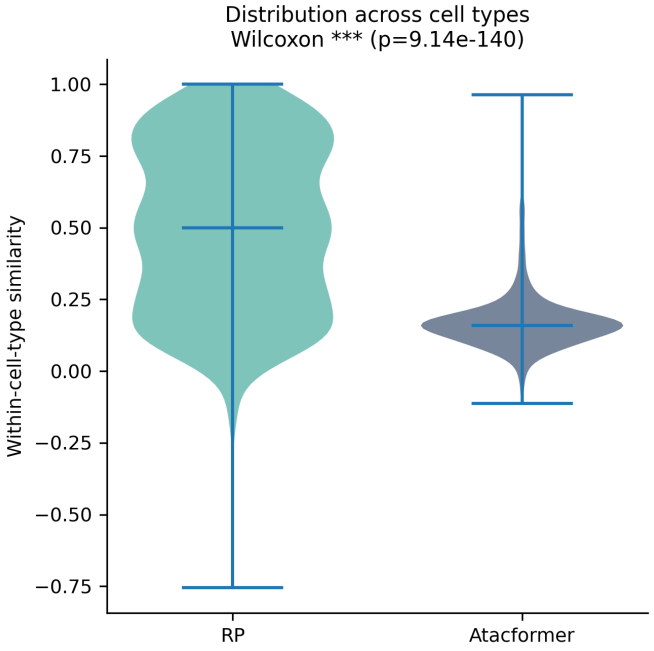


Fig. 12. All 32,245 experiments (contrast analysis). Distribution of within-cell-type similarity across cell types (Wilcoxon signed-rank test; RP median 0.50, Atacformer median 0.16). Within-cell-type similarity tracks the spread of peak counts within each cell type (Section III-A).

IV. Discussion

Our results establish quantitative bounds on the utility of experimental metadata as a proxy for ATAC-seq chromatin similarity, and show that these bounds depend critically on data quality. On the full ChIP-Atlas corpus, text embeddings appear to correlate moderately with Atacformer similarity ($r = 0.20$ to 0.30). This agreement is driven by low-peak experiments, whose promoter-dominated peak sets make them resemble one another in both chromatin representations, and by sample titles, which encode data provenance that covaries with library quality. Among well-covered experiments, text embeddings do not track Atacformer similarity at all.

What survives is a weak but consistent agreement between text and RP ($r = 0.06$ to 0.10 across experiments; 31 to 35% shared top-5 neighbors). RP neighborhoods are strongly organized by cell type, and text metadata recovers this coarse cell-type structure. Text metadata may therefore support coarse, cell-type-level retrieval, but it should not substitute for chromatin similarity in fine-grained applications.

Language models add little to this signal. In well-covered experiments, TF-IDF matches every LLM, and residual associations beyond TF-IDF are near zero for Atacformer and small for RP. The apparent advantage of PubMedBERT and SapBERT over MedCPT and the general encoder on the full dataset does not survive restriction to well-covered experiments and depends on corpus centering.

These findings have a methodological implication beyond text. Peak count was the strongest single determinant of similarity in both chromatin representations, and it alone accounted for the agreement between Atacformer and RP on the full dataset. Comparisons of chromatin embeddings across heterogeneous public repositories should therefore stratify by, or control for, peak count and related quality measures.

A central limitation is that Atacformer and RP are treated as proxies for chromatin biology without independent validation against external ground truth. Both are transforms of the same ChIP-Atlas peak calls, both are sensitive to peak count, and the Atacformer embeddings used here carry little cell-type information for well-covered experiments. All conclusions are bounded by how faithfully these representations capture the biology they proxy. Additional limitations include the absence of standard quality metrics (e.g., fraction of reads in peaks or TSS enrichment) in ChIP-Atlas, which limits us to peak count as a quality proxy; the pragmatic choice of a 5,000-peak threshold; the restriction to metadata fields available in ChIP-Atlas; and the absence of multimodal embeddings combining text and chromatin signals. Future work could re-embed experiments using all of their peaks, use quality-controlled reprocessed data, and test whether text encoders trained to predict chromatin similarity can narrow the gap.

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TABLE II

ALL 32,245 EXPERIMENTS (CONTRAST ANALYSIS). EXPERIMENT PAIRS WITH HIGH ATACFORMER SIMILARITY (TOP 5%) AND LOW TF-IDF SIMILARITY (BELOW MEDIAN). THE LEADING PAIR (NCI-H1299 / CD4+ T CELLS) CONSISTS OF TWO LOW-PEAK EXPERIMENTS (1,181 AND 1,272 PEAKS; CORPUS MEDIAN 6,287). "(UNLABELED)" DENOTES EXPERIMENTS LACKING A CELL-TYPE ANNOTATION IN CHIP-ATLAS.

Cell type A	Cell type B	Atac.	TFIDF
NCI-H1299	CD4+ T cells	0.374	−0.196
iPSC derived cells	PC-9	0.361	−0.176
NCI-H1648	iMeLCs	0.312	−0.214
(unlabeled)	CD4+ T cells	0.398	−0.147
NCI-H1648	iPSC neural cells	0.307	−0.204
(unlabeled)	IMR-90	0.309	−0.188
(unlabeled)	MV-4-11	0.331	−0.156
(unlabeled)	Neural crest cells	0.310	−0.179
iPSC derived cells	Jurkat	0.318	−0.164
(unlabeled)	DND-41	0.316	−0.161

TABLE III

ALL 32,245 EXPERIMENTS (CONTRAST ANALYSIS). EXPERIMENT PAIRS WITH HIGH RP SIMILARITY AND LOW TF-IDF SIMILARITY, SELECTED ANALOGOUSLY TO TABLE II. EVERY EXPERIMENT IN THIS TABLE HAS $\leq$1,438 PEAKS, I.E., THESE PAIRS REFLECT SHARED PROMOTER SIGNAL IN LOW-PEAK EXPERIMENTS RATHER THAN SHARED BIOLOGY.

Cell type A	Cell type B	RP	TFIDF
iPSC derived cells	K-562	0.885	−0.221
NCI-H2347	PBMC	0.884	−0.181
RERF-LC-Ad1	XG-6	0.851	−0.224
(unlabeled)	PKC19	0.901	−0.171
iPSC derived cells	T-47D	0.874	−0.185
(unlabeled)	B-cell leukemia	0.891	−0.170
NCI-H1648	B cells	0.859	−0.181
NCI-H1703	CD4+ T cells	0.851	−0.186
NCI-H1299	CD4+ T cells	0.833	−0.196
(unlabeled)	ES cells	0.916	−0.154